作业二

1. 证明语言 $L = \{u^{2n}v^{3n}r^{n} | n \geq 0\}$不是正则语言。

2. 设字母表为$\Sigma = \{a\}$，证明如下语言

$$L = \{a^{n} \in \Sigma^{*} \mid n \geq 2, \text{是一个素数}\}$$

不是正则的

3. 构造下推自动机接收如下上下文无关文法定义的语言。

$$S \rightarrow aSbb|abb$$

4. 设计语言 $L = \{a^{i}b^{j}c^{k} \mid i = 2j \text{ 或 } j = 2k\}$的上下文无关文法，并将其转化为下推自动机 PDA。

5. 构造一个下推自动机接收如下语言：

$$L = \{a^{n}b^{n+m}c^{m} \mid n \geq 0, m \geq 1\}，\text{其中} \sum = \{a, b, c\}$$

6. 给定一个下推自动机 $P = (\{q_0, q_1\}, \{0,1\}, \{X, z_0\}, \delta, q_0, z_0, \emptyset)$，其中

$$\delta(q_0, 0, z_0) = \{(q_0, Xz_0)\}$$

$$\delta(q_0, 0, X) = \{(q_0, XX)\}$$

$$\delta(q_0, 1, X) = \{(q_1, \varepsilon)\}$$

$$\delta(q_1, 1, X) = \{(q_1, \varepsilon)\}$$

$$\delta(q_1, \varepsilon, X) = \{(q_0, \varepsilon)\}$$

$$\delta(q_1, \varepsilon, z_0) = \{(q_0, \varepsilon)\}$$

请将 P 转化为 CFG。

第 1 题解答：

用反证法证明：

假设 L 是正则语言，从 L 中取字符串 $w = u^{2N}v^{3N}r^{N}$，

显然 $s \in L$ 并且$|S| = 6N \geq N$

根据泵引理，w 可以被写成 $w = xyz$，满足条件

1. $xy^{n}z \in L$ 对于任何 $n \geq 0$。
2. $|y| > 0$
3. $|xy| \leq N$

由于 w 的前 N 个字母都是 u，那么结合第三个条件，可以推出结论：x，y 只由 u 组成，所以 x，y，z 可以写为：

$$x = u^{j}, \text{对于 } j \geq 0$$

$$y = u^{k}, \text{对于 } k \geq 1$$

$$z = u^{m+N}v^{3N}r^{N}, \text{对于 } m \geq 0$$

同时 j, k, m 满足 $j + k + m + N = 2N$，即 $j + k + m = N$，所以 $j + k \leq N$。

根据上述的第一个条件，$xy^{2}z \in L$，而

$$xy^{2}z = u^{j}u^{k}u^{k}u^{m+N}v^{3N}r^{N} = u^{2N+k}v^{3N}r^{N}$$

由于 $k \geq 1$，上述的 $xy^{2}z$ 一定不属于 L，所以推出矛盾

因此，得证 $L = \{u^{2n}v^{3n}r^{n} | n \geq 0\}$不是正则语言。

第 2 题解答：

(a) Suppose L is regular and m is given. Let p be the smallest prime number such that $p \geq m$. Then we pick $w = a^p$ in L. The string y must then be a^k and the pumped strings will be

$$w_i = a^{p+(i-1)k} \in L,$$

for $i = 0, 1, \ldots$ However, $w_{p+1} = a^{p+pk} = a^{p(1+k)} \notin L$. Therefore L is not regular.

第 3 题解答：

Convert the grammar into Greibach normal form.

$$\begin{aligned} S &\rightarrow aSSSA|\lambda, \\ A &\rightarrow aB, \\ B &\rightarrow b. \end{aligned}$$

Following the construction of Theorem 7.1, we get the solution

$$\begin{aligned} \delta(q_0, \lambda, z) &= \{(q_1, Sz)\}, \\ \delta(q_1, a, S) &= \{(q_1, SSSA)\}, \\ \delta(q_1, a, A) &= \{(q_1, B)\}, \\ \delta(q_1, \lambda, S) &= \{(q_1, \lambda)\}, \\ \delta(q_1, b, B) &= \{(q_1, \lambda)\}, \\ \delta(q_1, \lambda, z) &= \{(q_f, z)\}. \end{aligned}$$

with $F = \{q_f\}$

注：λ为ε

第 4 题解答：

上下文无关文法如下：

$$S \rightarrow PC \mid AQ$$

$$P \rightarrow aaP\ b \mid \epsilon$$

$$C \rightarrow cC \mid \epsilon$$

$$A \rightarrow aA \mid \epsilon$$

$$Q \rightarrow bbQc \mid \epsilon$$

PDA 如下：

$$P = (\{q\}, \{a, b, c\}, \{a, b, c, A, B, P, Q, S\}, \delta, q, S, ?),$$

('?'表示只有一个状态，无终结状态集)

δ如下：

$$\delta(q, \epsilon, S) = \{(q, P\ C), (q, AQ)\}.$$

$$\delta(q, \epsilon, P) = \{(q, aaP\ b), (q, \epsilon)\}.$$

$$\delta(q, \epsilon, C) = \{(q, cC), (q, \epsilon)\}.$$

$$\delta(q, \epsilon, A) = \{(q, aA), (q, \epsilon)\}.$$

$$\delta(q, \epsilon, Q) = \{(q, bbQc), (q, \epsilon)\}.$$

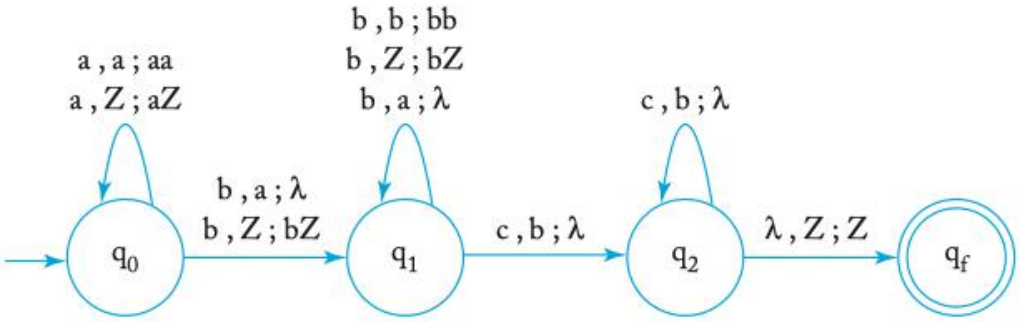
$$\delta(q, a, a) = \{(q, \epsilon)\}.$$

$$\delta(q, b, b) = \{(q, \epsilon)\}.$$

$$\delta(q, c, c) = \{(q, \epsilon)\}.$$

第 5 题解答：

Start with an initial z in the stack. Put a mark a on the stack when input symbol is an a, consume a mark a when input is a b. When all a's in the stack are consumed, put a mark b on the stack when input is a b. After input becomes c, eliminate a mark on the stack. The string will be accepted if the stack has only z left when the input is completed.



第 6 题解答：

令 A: $[q_0 Z_0 q_0]$

B: $[q_0 X q_0]$ $C[q_0 X q_1]$

E: $[q_1 Z_0 q_0]$ $F[q_1 X q_1]$

$G[q_1 X q_0]$

则 CFG：

$S \to A$

$A \to 0BA \mid 0CE$

$B \to 0BB \mid 0CG$

$C \to 0BC \mid 0CF \mid 1$

$E \to \varepsilon$

$F \to 1$

$G \to \varepsilon$

化简：$S \to 0BS \mid 0C$

$B \to 0BB \mid 0C$

$C \to 0BC \mid 0C1 \mid 1$

$S \to [q_0 Z_0 q_0]$

~~$S \to [q_0 Z_0 q_1]$~~

$[q_0 Z_0 q_0] \to 0[q_0 X q_0][q_0 Z_0 q_0]$

$[q_0 Z_0 q_0] \to 0[q_0 X q_1][q_1 Z_0 q_0]$

~~$[q_0 Z_0 q_1] \to 0[q_0 X q_0][q_0 Z_0 q_1]$~~

~~$[q_0 Z_0 q_1] \to 0[q_0 X q_1][q_1 Z_0 q_1]$~~

$[q_0 X q_0] \to 0[q_0 X q_0][q_0 X q_0]$

$[q_0 X q_0] \to 0[q_0 X q_1][q_1 X q_0]$

$[q_0 X q_1] \to 0[q_0 X q_0][q_0 X q_1]$

$[q_0 X q_1] \to 0[q_0 X q_1][q_1 X q_1]$

$[q_0 X q_1] \to 1$

$[q_1 Z_0 q_0] \to \varepsilon$

$[q_1 X q_1] \to 1$